

Homework 7

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1. List all the isomorphism classes of abelian groups of order 5400. Provide both the elementary divisor decomposition and the invariant factor decomposition of each.

Proof. The prime factorization of 5400 is $2^3 \cdot 3^3 \cdot 5^2$.

Invariant factor decomposition	Elementary divisor decomposition
$\mathbb{Z}_{5400}$	$\mathbb{Z}_{25} \times \mathbb{Z}_{27} \times \mathbb{Z}_8$
$\mathbb{Z}_{2700} \times \mathbb{Z}_2$	$(\mathbb{Z}_{25} \times \mathbb{Z}_{27} \times \mathbb{Z}_4) \times \mathbb{Z}_2$
$\mathbb{Z}_{1800} \times \mathbb{Z}_3$	$(\mathbb{Z}_{25} \times \mathbb{Z}_9 \times \mathbb{Z}_8) \times \mathbb{Z}_3$
$\mathbb{Z}_{1350} \times \mathbb{Z}_2 \times \mathbb{Z}_2$	$(\mathbb{Z}_{25} \times \mathbb{Z}_{27} \times \mathbb{Z}_2) \times \mathbb{Z}_2 \times \mathbb{Z}_2$
$\mathbb{Z}_{1080} \times \mathbb{Z}_5$	$(\mathbb{Z}_5 \times \mathbb{Z}_{27} \times \mathbb{Z}_8) \times \mathbb{Z}_5$
$\mathbb{Z}_{900} \times \mathbb{Z}_6$	$(\mathbb{Z}_{25} \times \mathbb{Z}_9 \times \mathbb{Z}_4) \times (\mathbb{Z}_3 \times \mathbb{Z}_2)$
$\mathbb{Z}_{600} \times \mathbb{Z}_3 \times \mathbb{Z}_3$	$(\mathbb{Z}_{25} \times \mathbb{Z}_3 \times \mathbb{Z}_8) \times \mathbb{Z}_3 \times \mathbb{Z}_3$
$\mathbb{Z}_{540} \times \mathbb{Z}_{10}$	$(\mathbb{Z}_5 \times \mathbb{Z}_{27} \times \mathbb{Z}_4) \times (\mathbb{Z}_5 \times \mathbb{Z}_2)$
$\mathbb{Z}_{450} \times \mathbb{Z}_6 \times \mathbb{Z}_2$	$(\mathbb{Z}_{25} \times \mathbb{Z}_9 \times \mathbb{Z}_2) \times (\mathbb{Z}_3 \times \mathbb{Z}_2) \times \mathbb{Z}_2$
$\mathbb{Z}_{360} \times \mathbb{Z}_{15}$	$(\mathbb{Z}_5 \times \mathbb{Z}_9 \times \mathbb{Z}_8) \times (\mathbb{Z}_5 \times \mathbb{Z}_3)$
$\mathbb{Z}_{300} \times \mathbb{Z}_6 \times \mathbb{Z}_3$	$(\mathbb{Z}_{25} \times \mathbb{Z}_3 \times \mathbb{Z}_4) \times (\mathbb{Z}_3 \times \mathbb{Z}_2) \times \mathbb{Z}_3$
$\mathbb{Z}_{270} \times \mathbb{Z}_{10} \times \mathbb{Z}_2$	$(\mathbb{Z}_5 \times \mathbb{Z}_{27} \times \mathbb{Z}_2) \times (\mathbb{Z}_5 \times \mathbb{Z}_2) \times \mathbb{Z}_2$
$\mathbb{Z}_{180} \times \mathbb{Z}_{30}$	$(\mathbb{Z}_5 \times \mathbb{Z}_9 \times \mathbb{Z}_4) \times (\mathbb{Z}_5 \times \mathbb{Z}_3 \times \mathbb{Z}_2)$
$\mathbb{Z}_{150} \times \mathbb{Z}_6 \times \mathbb{Z}_6$	$(\mathbb{Z}_{25} \times \mathbb{Z}_3 \times \mathbb{Z}_2) \times (\mathbb{Z}_3 \times \mathbb{Z}_2) \times (\mathbb{Z}_3 \times \mathbb{Z}_2)$
$\mathbb{Z}_{120} \times \mathbb{Z}_{15} \times \mathbb{Z}_3$	$(\mathbb{Z}_5 \times \mathbb{Z}_3 \times \mathbb{Z}_8) \times (\mathbb{Z}_5 \times \mathbb{Z}_3) \times \mathbb{Z}_3$
$\mathbb{Z}_{90} \times \mathbb{Z}_{30} \times \mathbb{Z}_2$	$(\mathbb{Z}_5 \times \mathbb{Z}_9 \times \mathbb{Z}_2) \times (\mathbb{Z}_5 \times \mathbb{Z}_3 \times \mathbb{Z}_2) \times \mathbb{Z}_2$
$\mathbb{Z}_{60} \times \mathbb{Z}_{30} \times \mathbb{Z}_3$	$(\mathbb{Z}_5 \times \mathbb{Z}_3 \times \mathbb{Z}_4) \times (\mathbb{Z}_5 \times \mathbb{Z}_3 \times \mathbb{Z}_2) \times \mathbb{Z}_3$
$\mathbb{Z}_{30} \times \mathbb{Z}_{30} \times \mathbb{Z}_6$	$(\mathbb{Z}_5 \times \mathbb{Z}_3 \times \mathbb{Z}_2) \times (\mathbb{Z}_5 \times \mathbb{Z}_3 \times \mathbb{Z}_2) \times (\mathbb{Z}_3 \times \mathbb{Z}_2)$

□

2. Let G be a group of order 225.

- (a) Show that $|Z(G)| \neq 1$.
- (b) Prove that if the Sylow 5-subgroup is cyclic, then G is abelian.
- (c) Find presentations for two non-isomorphic nonabelian groups of order 225.
- (d) Generalize part (a) to a group of order p^2q^2 for primes $p < q$. What hypotheses on p and q do you need for your argument from part (a) to hold?

Proof. Note that $225 = 3^2 \cdot 5^2$. □

3. Let H and K be groups, and let $\varphi: K \rightarrow \text{Aut}(H)$ be a group homomorphism. Let G be the set(not the group) $H \times K$; i.e. the set of ordered pairs (h, k) such that $h \in H$ and $k \in K$. For each $k \in K$, $\varphi(k) \in \text{Aut}(H)$, so $\varphi(k)(h) \in H$.

Define the product on G by

$$(h_1, k_1)(h_2, k_2) = (h_1\varphi(k_1)(h_2), k_1k_2).$$

- (a) Prove that this operation turns G into a group. This group is called the semidirect product of H and K with respect to φ , and is denoted $H \rtimes_{\varphi} K$.
 - (b) Show that the map $\varphi: G \rightarrow H \times K$ defined by $\varphi(h, k) = (h, k)$ is a group homomorphism if and only if $\varphi: K \rightarrow \text{Aut}(H)$ is the trivial homomorphism; i.e. $\varphi(k) = \text{id}_H$ for all $k \in K$. Note that this means in this case, $G \cong H \times K$.
 - (c) Prove that $\{(h, 1) \mid h \in H\}$ and $\{(1, k) \mid k \in K\}$ are subgroups of G , with the first isomorphic to H and the second isomorphic to K . In this way, we identify H and K with these subsets of G . So from now on, H will either mean H or the first subgroup mentioned above; similarly for K .
 - (d) Prove that H is normal in G and that $H \cap K = \{1\}$.
 - (e) Show that under the identification of $h \in H$ with $(h, 1) \in G$ and $k \in K$ with $(1, k) \in G$, the equality $khk^{-1} = \varphi(k)(h)$ holds for all $h \in H$ and $k \in K$. This means that we have realized the action of K on H via the homomorphism $\varphi: K \rightarrow \text{Aut}(H)$ as the conjugation action of the subgroup K on the normal subgroup H .
 - (f) Show that K is normal in G if and only if $G \cong H \times K$.
- 4.** Suppose H and K are subgroups of a group G with H normal in G and $H \cap K = \{1\}$. Let $\varphi: K \rightarrow \text{Aut}(H)$ be the homomorphism given by $\varphi(k)(h) = khk^{-1}$. Prove that $HK \cong H \rtimes_{\varphi} K$.
- 5.** (Optional, but very useful!) Let K be a finite cyclic group, H an arbitrary group, and let $\varphi_1, \varphi_2: K \rightarrow \text{Aut}(H)$ be group homomorphisms.

- (a) Prove that if $\varphi_1(K)$ and $\varphi_2(K)$ are conjugate in $\text{Aut}(H)$, then there exists $\sigma \in \text{Aut}(H)$ and $a \in \mathbb{Z}$ such that $\sigma\varphi_1(k)\sigma^{-1} = \varphi_2(k^a)$ for all $k \in K$.
- (b) For the next part, we need a number-theoretic lemma, which we will simply assume:

Lemma: Suppose that a, m, n are integers such that $m \mid n$ and $\gcd(a, m) = 1$. Then there exists an integer a' such that $a' \equiv a \pmod{m}$ and $\gcd(a', n) = 1$.

Use the lemma to prove that the a in part (a) can be chosen to be relatively prime to $|K|$.

- (c) With the σ and a as in the previous parts, prove that the following map is a group homomorphism:

$$\begin{aligned}\Phi : H \rtimes_{\varphi_1} K &\rightarrow H \rtimes_{\varphi_2} K \\ (h, k) &\mapsto (\sigma(h), k^a)\end{aligned}$$

- (d) Prove¹ Φ is an isomorphism by constructing a two-sided inverse.

- 6.** Find a nonabelian group of order 27 expressed as a semidirect product. Use this information to find a presentation of your group.

Proof. Consider $\mathbb{Z}_3 \rtimes \mathbb{Z}_9$. This has the presentation

$$\langle x, y : x^3 = y^9 = e, yxy^{-1} = x^{-1} \rangle.$$

It is nonabelian since $yx = x^{-1}y = x^2y$. □

- 7.** Determine (with justification) all groups of order 1805 up to isomorphism, and find² a presentation of each.

Proof. Since $1805 = 5 \cdot 19^2$, we only have two possible invariant factor decompositions: $\mathbb{Z}_{1805}$ and $\mathbb{Z}_{95} \times \mathbb{Z}_{19}$. This is due to the restrictions on divisibility ($|\mathbb{Z}_{i+1}|$ divides $|\mathbb{Z}_i|$) together with the fact that their orders must multiply to 1805.

A presentation of $\mathbb{Z}_{1805}$ is easy; just take

$$\langle x : x^{1805} = e \rangle.$$

For the other, notice that $\mathbb{Z}_{95} \cong \mathbb{Z}_{19} \times \mathbb{Z}_5$ since 19 and 5 are coprime. Thus the group becomes $\mathbb{Z}_{19} \times \mathbb{Z}_{19} \times \mathbb{Z}_5$, which has the easy presentation

$$\langle x, y, z : x^{19} = y^{19} = z^5 = e \rangle.$$

□

¹The integer a and the order of K are relatively prime, so there must exist a b such that $ab \equiv 1$ modulo $|K|$. Use this to show $\sigma^{-1}\varphi_2(k)\sigma = \varphi_1(k^b)$ for all $k \in K$.

²The matrix $\begin{pmatrix} 4 & 1 \\ -1 & 0 \end{pmatrix}$ has order five in $\text{GL}_2(\mathbb{F}_{19})$.